

Lecture 12

Hamiltonian reduction: where the Lax pair comes from

Objectives. Moser's matrix was a rabbit. Here it stops being one. A free particle moving in a straight line through the $N \times N$ Hermitian matrices has eigenvalues that repel — and are Calogero–Moser particles. The machinery that says why is built only for this example: conjugation, its momentum map (a commutator), a level set, a gauge fixing. Reducing at momentum zero gives free particles, the wrong model; at one non-zero value it gives Calogero–Moser, with Moser's L the gauge-fixed momentum and Moser's M the velocity of the gauge transformation.

12.1 Free motion in disguise

Start with the picture (Figure 12.1). Take two 3×3 Hermitian matrices X_0, P_0 and move along the straight line

$$X(t) = X_0 + t P_0. \quad (12.1)$$

Nothing could be freer. Yet the three eigenvalues of $X(t)$, straight lines far in the past and future, do not cross: they approach, slow down and turn away, as if they repelled. For the right X_0, P_0 they obey exactly the rational Calogero–Moser equations of Lecture 11, and Moser's Lax pair is what is left of the free motion once we look only at the eigenvalues.

Example 12.1 (Two eigenvalues and the centrifugal barrier). For $N = 2$ every Hermitian matrix is $X = q_c \mathbb{1} + \mathbf{x} \cdot \boldsymbol{\sigma}$, with $q_c \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^3$ and $\boldsymbol{\sigma}$ the Pauli matrices. Its eigenvalues are $q_c \pm |\mathbf{x}|$, so their gap is $s = 2|\mathbf{x}|$. Along (12.1), q_c moves uniformly and $\mathbf{x}(t) = \mathbf{x}_0 + t\mathbf{v}$ is a free particle in $\mathbb{R}^3$. Its distance from the origin feels the *centrifugal barrier*: with $\boldsymbol{\ell} = \mathbf{x} \times \mathbf{v}$ conserved, $|\mathbf{v}|^2 = \dot{r}^2 + |\boldsymbol{\ell}|^2 / r^2$, $r = |\mathbf{x}|$. In terms of the gap,

$$|\mathbf{v}|^2 = \frac{1}{4}\dot{s}^2 + \frac{4|\boldsymbol{\ell}|^2}{s^2},$$

which is the relative energy of two Calogero–Moser particles, $\frac{1}{4}\dot{s}^2 + g^2/s^2$, with $g = 2|\boldsymbol{\ell}|$. The eigenvalues repel because a straight line that misses the origin cannot reach it (Figure 12.2); they would collide only if $\boldsymbol{\ell} = 0$, a line through the origin.

For two particles the story is familiar: fix the angular momentum, forget the direction, and the radial motion is an inverse-square problem. For N the rotations become the unitary group $U(N)$ acting on matrices, the angular momentum becomes the commutator $[X, P]$, and “fix it” needs one idea that $N = 2$ hides: *which* value to fix it at.

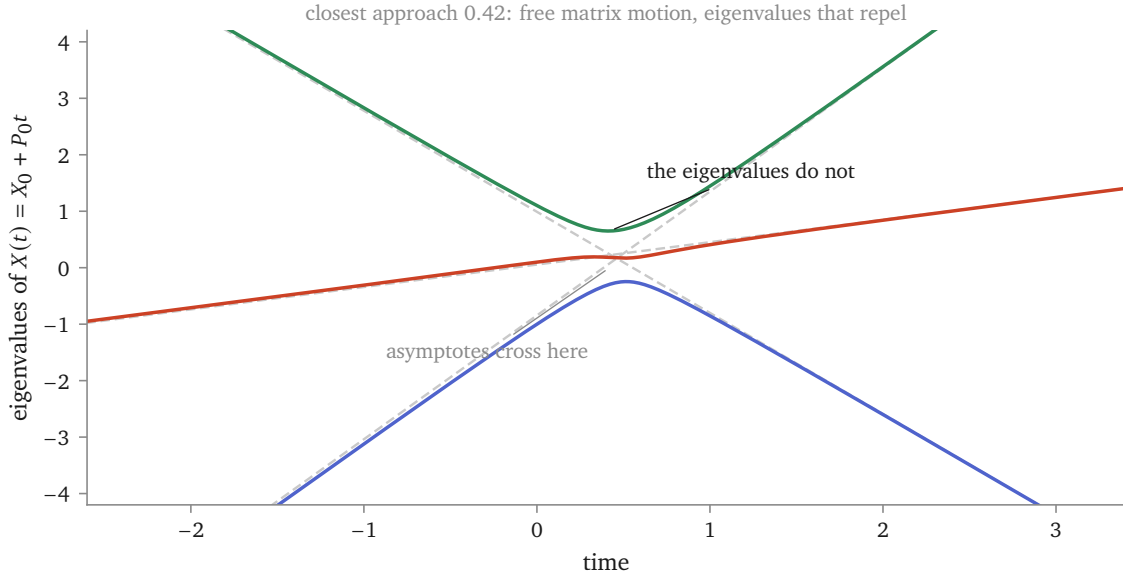


Figure 12.1: The eigenvalues of $X_0 + tP_0$, a straight line in the space of 3×3 Hermitian matrices, with $X_0 = \text{diag}(q)$ and P_0 Moser's L of (11.2) at $g = 0.6$. The asymptotes (dashed) cross; the eigenvalues do not. (scripts/figures/cm_eigenvalue_repulsion.py integrates Calogero–Moser from the same data and asserts agreement to 10^{-7} .)

12.2 Group actions, orbits and stabilisers

§ 4.4 used an action of $\mathbb{R}^n$, its orbits and its stabilisers (Definition 4.9). The same words, for any group:

Definition 12.2 (Group action, orbit, stabiliser). An action of a group G on a set S assigns to each $U \in G$ a map $\Phi_U : S \rightarrow S$ with $\Phi_1 = \text{id}$ and $\Phi_U \circ \Phi_V = \Phi_{UV}$. The orbit of $s \in S$ is $\{\Phi_U(s) : U \in G\}$; its stabiliser is $G_s = \{U : \Phi_U(s) = s\}$, a subgroup.

Orbits partition S . Points on one orbit have conjugate stabilisers, $G_{\Phi_V(s)} = VG_sV^{-1}$; for the abelian group of Lemma 4.10 that made them equal.

The example. $\text{Herm}(N)$, the $N \times N$ Hermitian matrices, is a real vector space of dimension N^2 (N real diagonal entries, $N(N-1)/2$ complex ones above the diagonal), and

$$\langle A, B \rangle = \text{tr}(AB) \quad (12.2)$$

is a Euclidean inner product on it: $\text{tr}(A^2) = \sum_{j,k} |A_{jk}|^2$. The unitary group $U(N)$ acts by conjugation, $\Phi_U(X) = UXU^\dagger$, and preserves (12.2): $\text{tr}(UAU^\dagger UB U^\dagger) = \text{tr}(AB)$. So conjugation is a group of rotations of the Euclidean space $\text{Herm}(N) \cong \mathbb{R}^{N^2}$. Scalars $U = e^{i\theta} \mathbb{1}$ act trivially; this will matter every time we count.

Proposition 12.3 (Orbits of conjugation). (a) The orbit of X is the set of Hermitian matrices with the same eigenvalues, counted with multiplicity. Each orbit contains exactly one diagonal matrix $Q = \text{diag}(q_1, \dots, q_N)$ with $q_1 \leq \dots \leq q_N$.

(b) Write $\mathfrak{u}(N)$ for the anti-Hermitian matrices, the velocities $\dot{U}(0)$ of curves in $U(N)$ through 1. The velocities of curves in the orbit through X are the matrices $[A, X]$, $A \in \mathfrak{u}(N)$.

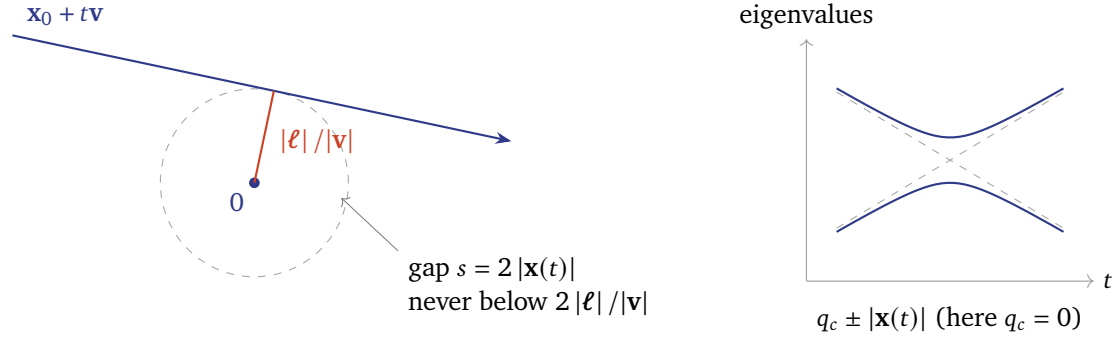


Figure 12.2: Free motion in matrix space, $N = 2$. Left: the traceless part $\mathbf{x}$ of $X(t)$ moves on a straight line in $\mathbb{R}^3$; the closest it gets to the origin is the impact parameter $|\ell|/|\mathbf{v}|$. Right: the two eigenvalues are $\pm |\mathbf{x}(t)|$, so they approach, turn and separate — the centrifugal barrier, seen as a repulsion.

(c) If $X = Q$ is regular — all q_j distinct — then $[A, Q] = 0$ exactly for diagonal A , the stabiliser $U(N)_Q$ is the group T^N of diagonal unitaries, and the orbit has dimension $N^2 - N$.

Proof. (a) is the spectral theorem: $X = UQU^\dagger$ with U unitary and Q real diagonal, and the order of the q_j can be chosen by permuting the columns of U . (b) If $U(s)$ is a curve in $U(N)$ with $U(0) = \mathbb{1}$ and $\dot{U}(0) = A$, then $U^\dagger U = \mathbb{1}$ gives $A^\dagger + A = 0$, and $\frac{d}{ds} U(s)XU(s)^\dagger \Big|_0 = AX + XA^\dagger = [A, X]$; conversely $U(s) = e^{sA}$ realises any A . (c) Entry by entry, $[A, Q]_{jk} = (q_k - q_j)A_{jk}$, which vanishes for all $j \neq k$ exactly when A is diagonal. A unitary U with $UQ = QU$ maps each eigenline of Q to itself, so is diagonal. The velocities $[A, Q]$ fill the $N^2 - N$ real dimensions of the off-diagonal part, so that is the orbit's dimension (its tangent space at Q). ■

12.3 The momentum map is a commutator

The phase space

Throughout, $N \geq 2$. The phase space of a free particle in $\text{Herm}(N)$ is $\mathcal{M} = \text{Herm}(N) \times \text{Herm}(N)$, points (X, P) : a position matrix and a momentum matrix. (P here is a matrix of momenta; Lecture 8's permutation operator does not appear in this lecture. And X is a matrix, not a vector field: Hamiltonian vector fields keep their subscript, X_f .) Choose an orthonormal basis $T_1, \dots, T_{N^2}$ of $\text{Herm}(N)$ for (12.2) — for instance E_{jj} , $(E_{jk} + E_{kj})/\sqrt{2}$ and $i(E_{jk} - E_{kj})/\sqrt{2}$ for $j < k$ — and write $X = \sum_a x_a T_a$, $P = \sum_a p_a T_a$. We declare (x_a, p_a) canonical:

$$\omega = \sum_{a=1}^{N^2} dx_a \wedge dp_a =: \text{tr}(dX \wedge dP), \quad H = \frac{1}{2} \text{tr} P^2 = \frac{1}{2} \sum_a p_a^2. \quad (12.3)$$

So $\mathcal{M}$ is just $\mathbb{R}^{2N^2}$ with Lecture 1's bracket, and H is a free particle: $\dot{X} = P$, $\dot{P} = 0$, which is (12.1). For a function $f(X, P)$ write $\partial f / \partial X = \sum_a (\partial f / \partial x_a) T_a$, a Hermitian matrix; then $\{f, h\} = \text{tr} \left(\frac{\partial f}{\partial X} \frac{\partial h}{\partial P} - \frac{\partial f}{\partial P} \frac{\partial h}{\partial X} \right)$, and a linear function has the obvious gradient: $\partial \text{tr}(BX) / \partial X = B$ for Hermitian B .

$U(N)$ acts on $\mathcal{M}$ by conjugating both matrices, $\Phi_U(X, P) = (UXU^\dagger, UPU^\dagger)$. In the coordinates (x_a, p_a) this is the same rotation O of $\mathbb{R}^{2N^2}$ applied to x and to p , and any such map is a symplectomorphism (Definition 3.13): $\sum dx_a \wedge dp_a$ is the bilinear form $x \cdot p' - p \cdot x'$ on pairs of tangent vectors, which O preserves. And H is invariant: $\text{tr}(UPU^\dagger)^2 = \text{tr} P^2$.

Momentum maps

A one-parameter subgroup $U(s) = e^{sA}$, $A \in \mathfrak{u}(N)$, moves points of $\mathcal{M}$ with velocity $([A, X], [A, P])$ at $s = 0$ (Proposition 12.3(b)). Lecture 1's Noether principle says a function whose flow is a symmetry of H is conserved; a momentum map is the bookkeeping device that produces such a function for every A at once.

Definition 12.4 (Momentum map). A momentum map for the action of $U(N)$ on $\mathcal{M}$ is a function J from $\mathcal{M}$ to $\mathfrak{u}(N)$ such that, for every $A \in \mathfrak{u}(N)$, the real function $J_A = \text{tr}(AJ)$ has Hamiltonian vector field equal to the velocity field of e^{sA} :

$$X_{J_A}X = [A, X], \quad X_{J_A}P = [A, P],$$

the vector field acting on the matrix coordinates entry by entry.

The literature writes μ , which here is the spectral parameter. The trace pairing replaces the usual map into the *dual* of the Lie algebra: $\text{tr}(AB)$ is real and non-degenerate on $\mathfrak{u}(N)$, so one matrix $J(X, P)$ records all the J_A . For translations of a particle in $\mathbb{R}^3$ the momentum map is the momentum; for rotations it is the angular momentum (Example 3.7).

Proposition 12.5 (The momentum map is a commutator). $J(X, P) = [X, P]$ is a momentum map for conjugation.

Proof. $[X, P]^\dagger = [P, X] = -[X, P]$, so J is anti-Hermitian, and $J_A = \text{tr}(A[X, P])$ is real, since the trace of a product of two anti-Hermitian matrices is. By cyclicity $J_A = \text{tr}(P[A, X]) = \text{tr}(X[P, A])$, and $[A, X]$, $[P, A]$ are Hermitian, so the linear-gradient rule gives $\partial J_A / \partial P = [A, X]$ and $\partial J_A / \partial X = [P, A]$. With $X_f h = \{h, f\}$:

$$X_{J_A}X = \{X, J_A\} = \frac{\partial J_A}{\partial P} = [A, X], \quad X_{J_A}P = \{P, J_A\} = -\frac{\partial J_A}{\partial X} = [A, P]. \quad \blacksquare$$

For $N = 2$ the traceless part of J is $2i(\mathbf{x} \times \mathbf{v}) \cdot \boldsymbol{\sigma}$: the angular momentum of Example 12.1, as it should be.

Corollary 12.6 (Conserved and equivariant). (a) J is constant along the flow of any $U(N)$ -invariant H ; for $H = \frac{1}{2} \text{tr} P^2$, directly, $\frac{d}{dt} [X, P] = [P, P] + [X, 0] = 0$.
(b) $J(UXU^\dagger, UPU^\dagger) = UJ(X, P)U^\dagger$.

Proof. (a) $\{J_A, H\} = -X_{J_A}H = -\frac{d}{ds}H(\Phi_{e^{sA}})|_0 = 0$, Noether as in Lecture 1. (b) is $[UXU^\dagger, UPU^\dagger] = U[X, P]U^\dagger$. $\blacksquare$

So a trajectory that starts on a level set $J^{-1}(\xi) = \{(X, P) : [X, P] = \xi\}$ stays on it: $J = \xi$ is a *constraint* imposed once and for all. The general theorem behind the rest of the lecture is boxed here; below, it supplies only the reduced symplectic form and its brackets, and everything else is proved on the example.

Theorem 12.7 (Marsden–Weinstein reduction · cited). Let a compact group G act on $(\mathcal{M}, \omega)$ by symplectomorphisms with an equivariant momentum map J , and let ξ be a value of J at which dJ is onto at every point of $J^{-1}(\xi)$ (a regular value), and on which the stabiliser G_ξ of ξ acts freely (a subgroup acting trivially on all of $\mathcal{M}$, like the scalars here, is divided out of G first). Then $\mathcal{M}_\xi = J^{-1}(\xi)/G_\xi$ is a manifold of dimension $\dim \mathcal{M} - \dim G - \dim G_\xi$, carrying a unique symplectic form ω_ξ whose pull-back to $J^{-1}(\xi)$ is the restriction of ω . A G -invariant Hamiltonian defines a function on $\mathcal{M}_\xi$, whose flow is the image of the original

flow, and Poisson brackets of G -invariant functions pass to $\mathcal{M}_\xi$. (Marsden and Weinstein 1974; da Silva, Thm 23.1 and §24.4.)

12.4 Level sets, the stabiliser, and the residual gauge group

Which level, and what still acts on it

$J = [X, P]$ is anti-Hermitian and traceless ($\text{tr}[X, P] = 0$), so a level ξ must be too. This matches the scalars acting trivially: the group that acts effectively is $\text{PU}(N) = \text{U}(N)/\text{U}(1)$, and through the trace pairing its Lie algebra is $\mathfrak{su}(N)$, the traceless anti-Hermitian matrices. From here on J is regarded as a map $\mathcal{M} \rightarrow \mathfrak{su}(N)$, of dimension $N^2 - 1$, and that is the J to which [Theorem 12.7](#) is applied, with $G = \text{PU}(N)$. Fix a level ξ . By [Corollary 12.6\(b\)](#), Φ_U carries $J^{-1}(\xi)$ onto $J^{-1}(U\xi U^\dagger)$. For $\xi \neq 0$ that is generally a *different* level set: $\text{U}(N)$ does not act on $J^{-1}(\xi)$, so $J^{-1}(\xi)/\text{U}(N)$ does not make sense. What does act is the *stabiliser*

$$\text{U}(N)_\xi = \{U \in \text{U}(N) : U\xi U^\dagger = \xi\},$$

and $J^{-1}(\xi)/\text{U}(N)_\xi$ is what replaces it. Physicists call $\text{U}(N)_\xi$ the *residual gauge group*: two points of $J^{-1}(\xi)$ related by one of its elements are the same physical state.

At $\xi = 0$ the stabiliser is all of $\text{U}(N)$. At the value we shall need,

$$\xi = \text{ig}(ee^T - 1), \quad e = (1, \dots, 1)^T, \quad \text{that is} \quad \xi_{jk} = \text{ig}(1 - \delta_{jk}), \quad g > 0, \quad (12.4)$$

it is much smaller. (ξ is anti-Hermitian and traceless, as it must be.)

Proposition 12.8 (The stabiliser and orbit of ξ). (a) $U \in \text{U}(N)_\xi$ if and only if $Ue = e^{i\theta}e$ for some θ . So $\text{U}(N)_\xi \cong \text{U}(1) \times \text{U}(N-1)$ (a phase on e , any unitary on $e^\perp$), of dimension $1 + (N-1)^2$; it contains the permutation matrices.
(b) *sketched* The orbit of ξ is $\{\text{ig}(ww^\dagger - 1) : w \in \mathbb{C}^N, |w|^2 = N\}$, of dimension $2N - 2$.

Proof. (a) $U\xi U^\dagger = \text{ig}(Ue(Ue)^\dagger - 1)$, which equals ξ exactly when $Ue(Ue)^\dagger = ee^T$, i.e. when Ue is a unit multiple of e . A permutation matrix fixes e . (b) $U\xi U^\dagger = \text{ig}(ww^\dagger - 1)$ with $w = Ue$, and $\text{U}(N)$ moves e to every vector of the same length. The dimension is $\dim \text{U}(N) - \dim \text{U}(N)_\xi = N^2 - (1 + (N-1)^2) = 2N - 2$: as in [Proposition 12.3\(b\)](#), the orbit's velocities are the $[A, \xi]$, and the A with $[A, \xi] = 0$ are the stabiliser's. ■

The constraint forbids collisions

Lemma 12.9 (No collisions). Let $[X, P] = \xi$, and diagonalise, $X = VQV^\dagger$. Then $w = V^\dagger e$ has $|w_j| = 1$ for every j , and X is regular: its eigenvalues are distinct.

Proof. Conjugating by $V^\dagger$, $[Q, \tilde{P}] = \text{ig}(ww^\dagger - 1)$ with $\tilde{P} = V^\dagger PV$. Entry by entry, $[Q, \tilde{P}]_{jk} = (q_j - q_k)\tilde{P}_{jk}$. On the diagonal the left side is 0, so $\text{ig}(|w_j|^2 - 1) = 0$. Off the diagonal, $(q_j - q_k)\tilde{P}_{jk} = \text{ig} w_j \bar{w}_k$, whose right side has modulus $g \neq 0$; so $q_j \neq q_k$. ■

This is the whole mechanism of [Figure 12.1](#) in one line: on $J^{-1}(\xi)$ the eigenvalues of X can never meet. Nothing like it holds at $\xi = 0$, where $X = 0$, P arbitrary, is allowed.

Dimensions

The constraint $[X, P] = \xi$ is $N^2 - 1$ real equations (the trace is automatic), independent on $J^{-1}(\xi)$: ξ is a regular value of $J : \mathcal{M} \rightarrow \mathfrak{su}(N)$, and $J^{-1}(\xi)$ is a manifold of dimension $N^2 + 1$. At $\xi = 0$ the torus T^N fixes a point $(Q, \text{diag}(p))$, $N - 1$ equations become dependent there, and the level set has dimension $N^2 + N$; at $(X, P) = (0, 0)$ it is not a manifold at all. 0 is not a regular value, and [Theorem 12.7](#) does not apply to it as stated. (Proofs in the reading at the end of the lecture.)

12.5 The reduction, done twice

The recipe is the same both times: use the residual gauge group to bring X to a normal form (gauge fixing), solve the constraint for what is left of P , and read off the Hamiltonian ([Figure 12.3](#)).

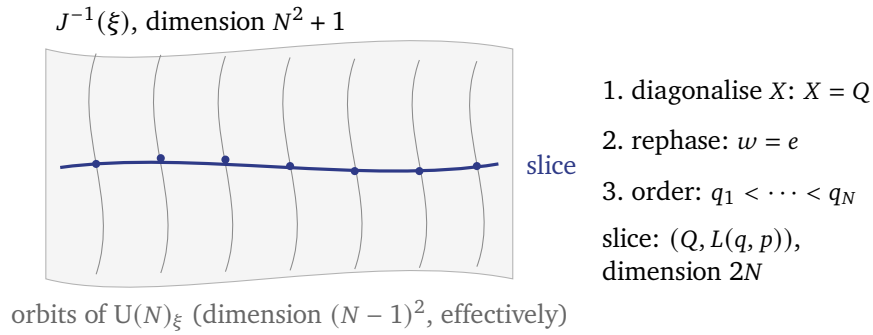


Figure 12.3: Gauge fixing. The level set is a union of orbits of the residual gauge group (grey); the slice (indigo) meets each exactly once, and is the reduced phase space.

At $\xi = 0$: the wrong model

Proposition 12.10 (Reduction at zero gives free particles). *Every point of $J^{-1}(0)$ with X regular is $U(N)$ -equivalent to $(Q, \text{diag}(p))$ with $q_1 < \dots < q_N$, and to exactly one such. On these points $H = \frac{1}{2} \sum_j p_j^2$: free particles.*

Proof. Conjugate X to Q ([Proposition 12.3\(a\)](#)); the constraint becomes $[Q, P]_{jk} = (q_j - q_k)P_{jk} = 0$, and $q_j \neq q_k$ forces $P_{jk} = 0$ for $j \neq k$. Uniqueness: a unitary carrying one such pair to another fixes the ordered Q , so is diagonal ([Proposition 12.3\(c\)](#)), and diagonal unitaries fix every diagonal P . ■

The dimension is right, $2N$; the Hamiltonian is wrong. In [Figure 12.1](#) this is a diagonal P_0 , whose eigenvalues $q_j + tp_j$ cross; in [Example 12.1](#), zero angular momentum, a line through the origin and no barrier.

At $\xi = \text{ig}(ee^T - 1)$: **Calogero–Moser**

Theorem 12.11 (Calogero–Moser by reduction (Kazhdan–Kostant–Sternberg)). (a)

Every point of $J^{-1}(\xi)$ is $U(N)_\xi$ -equivalent to exactly one point of the slice

$$\begin{aligned} X = Q = \text{diag}(q_1, \dots, q_N), \quad q_1 < \dots < q_N, \\ P = L(q, p), \quad L_{jk} = p_j \delta_{jk} + \frac{\text{ig}(1 - \delta_{jk})}{q_j - q_k}, \end{aligned} \quad (12.5)$$

which is Moser's L of (11.2) with $x = 1/u$.

(b) On the slice, $H = \frac{1}{2} \text{tr} L^2 = \frac{1}{2} \sum_j p_j^2 + g^2 \sum_{j < k} (q_j - q_k)^{-2}$, the rational Calogero–Moser Hamiltonian (11.1).

(c) The restriction of $\omega = \text{tr}(dX \wedge dP)$ to the slice is $\sum_j dq_j \wedge dp_j$. So the reduced phase space of Theorem 12.7 is the chamber $q_1 < \dots < q_N$ times $\mathbb{R}^N$, with its canonical form.

Proof. (a) Take (X, P) with $[X, P] = \xi$ and diagonalise, $X = VQV^\dagger$, eigenvalues increasing. By Lemma 12.9, $w = V^\dagger e$ has entries of modulus one, so $D = \text{diag}(w_1, \dots, w_N)$ is unitary. Put $U = VD$. Then $U^\dagger XU = D^\dagger QD = Q$, and $U^\dagger e = D^\dagger w = e$, so $U \in U(N)_\xi$ by Proposition 12.8(a). The pair $(Q, L) = (U^\dagger XU, U^\dagger PU)$ still satisfies $[Q, L] = U^\dagger \xi U = \xi$, that is $(q_j - q_k)L_{jk} = \text{ig}$ for $j \neq k$; so $L_{jk} = \text{ig}/(q_j - q_k)$, and the diagonal $p_j := L_{jj}$ is free. Uniqueness: if $U \in U(N)_\xi$ carries one slice point to another, it conjugates Q to an ordered diagonal matrix with the same eigenvalues, so fixes Q and is diagonal; a diagonal U with $Ue = e^{i\theta}e$ is the scalar $e^{i\theta}1$, which moves nothing.

(b) $\text{tr} L^2 = \sum_j p_j^2 + \sum_{j \neq k} L_{jk}L_{kj}$ and $L_{jk}L_{kj} = \text{ig} \cdot \text{ig}/((q_j - q_k)(q_k - q_j)) = g^2/(q_j - q_k)^2$.

(c) On the slice $dX = \text{diag}(dq_j)$ has no off-diagonal part, and in $\text{tr}(dX \wedge dP) = \sum_{j,k} dX_{jk} \wedge dP_{kj}$ only the terms $j = k$ survive: $\sum_j dq_j \wedge dL_{jj} = \sum_j dq_j \wedge dp_j$. By Theorem 12.7 the reduced form is the one whose pull-back is the restriction of ω ; the slice meets each gauge orbit once, so this is it. ■

Part (c) uses the cited theorem only for the *existence* of the reduced form. The dynamics can be followed without it, and this is the theorem the lecture was for.

Theorem 12.12 (The eigenvalues of free matrix motion). Let $(q(t), p(t))$ be the rational Calogero–Moser trajectory from $(q(0), p(0))$, $q_1(0) < \dots < q_N(0)$, and $Q(t) = \text{diag}(q(t))$. Then:

(a) $q_1(t) < \dots < q_N(t)$ are the eigenvalues of

$$X(t) = Q(0) + tL(0), \quad L(0) = L(q(0), p(0)), \quad (12.6)$$

for all $t \in \mathbb{R}$, and in particular the flow is complete;

(b) there are unitary $U(t) \in U(N)_\xi$, smooth in t , with $U(0) = 1$, $U^\dagger XU = Q(t)$ and $U^\dagger L(0)U = L(q(t), p(t))$;

(c) $L(t) = L(q(t), p(t))$ satisfies $\dot{L} = [M, L]$ with

$$M = -U^\dagger \dot{U}, \quad M_{jk} = \frac{\text{ig}}{(q_j - q_k)^2} \ (j \neq k), \quad M_{jj} = -\text{ig} \sum_{l \neq j} \frac{1}{(q_j - q_l)^2}, \quad (12.7)$$

which is Moser's M ((11.3)–(11.5)).

Proof. Run the argument backwards: start from the free motion (12.6), which is defined for all t and has $P(t) = L(0)$ and $[X(t), P] = [Q(0), L(0)] = \xi$ for all t (Corollary 12.6). By Lemma 12.9 the eigenvalues $\tilde{q}_1(t) < \dots < \tilde{q}_N(t)$ of $X(t)$ stay distinct, so the orthogonal projections $\Pi_j(t)$ onto the eigenlines are smooth in t (for a simple eigenvalue, $\Pi_j = \frac{1}{2\pi i} \oint (z - X)^{-1} dz$ around $\tilde{q}_j$ alone). Let $U(t)$ have columns $u_j = \Pi_j e$. In the notation of the lemma, $u_j = v_j w_j$ with v_j a unit eigenvector and $|w_j| = 1$, so U is unitary; and $u_j^\dagger e = |w_j|^2 = 1$, so $U^\dagger e = e$: this is the gauge of Theorem 12.11(a), now chosen smoothly, and $U(0) = \mathbb{1}$ since $X(0)$ is diagonal. So $U(t) \in U(N)_\xi$, $U^\dagger X U = \tilde{Q}(t)$, and $\tilde{L}(t) := U^\dagger L(0) U$ has the form (12.5) with some diagonal $\tilde{p}(t)$. Put $A = U^\dagger \dot{U}$, anti-Hermitian since $U^\dagger U = \mathbb{1}$.

Differentiate $\tilde{Q} = U^\dagger X U$, using $\dot{X} = L(0)$, $\dot{U}^\dagger = -A U^\dagger$ and $XU = U\tilde{Q}$:

$$\dot{\tilde{Q}} = -A\tilde{Q} + \tilde{L} + \tilde{Q}A = \tilde{L} + [\tilde{Q}, A].$$

$[\tilde{Q}, A]$ has zero diagonal, so the diagonal says $\dot{\tilde{q}}_j = \tilde{p}_j$. Off the diagonal the left side is 0, so $A_{jk} = -\tilde{L}_{jk}/(\tilde{q}_j - \tilde{q}_k) = -ig/(\tilde{q}_j - \tilde{q}_k)^2$.

The diagonal of A is the gauge condition. $U^\dagger e = e$ for all t , so $0 = \dot{U}^\dagger e = -A U^\dagger e = -Ae$: every row of A sums to zero, and $A_{jj} = -\sum_{l \neq j} A_{jl} = ig \sum_{l \neq j} (\tilde{q}_j - \tilde{q}_l)^{-2}$. So $M = -A$ is (12.7).

Differentiate $\tilde{L} = U^\dagger L(0) U$: $\dot{\tilde{L}} = -A\tilde{L} + \tilde{L}A = [M, \tilde{L}]$. Its diagonal, computed as in Theorem 11.1, is $\dot{\tilde{p}}_j = 2g^2 \sum_{l \neq j} (\tilde{q}_j - \tilde{q}_l)^{-3} = -\partial H / \partial q_j$.

So $(\tilde{q}, \tilde{p})$ solves Hamilton's equations for the Calogero–Moser Hamiltonian, for all t , from $(q(0), p(0))$. By uniqueness of solutions (Theorem 1.5) it is (q, p) , which proves (a), (b) and (c) at once. ■

Two promises of Lecture 11 are now kept. Theorem 11.8's sketch rested on (12.6), which is now proved; and the diagonal of Moser's M , which § 11.2 had to fix by solving a functional equation, is here nothing but the statement that the gauge transformation keeps e fixed. For the harmonic trap of § 11.6, replace H by $\frac{1}{2} \text{tr}(P^2 + \nu^2 X^2)$. It is still $U(N)$ -invariant, so J is still conserved; the matrix now oscillates, $\dot{X} = P$, $\dot{P} = -\nu^2 X$, and the same proof, with one extra term $-\nu^2 \tilde{q}_j$ in $\dot{\tilde{p}}_j$, gives:

Corollary 12.13 (The trap). *The trapped rational Calogero–Moser positions are the eigenvalues of $Q(0) \cos \nu t + \nu^{-1} L(0) \sin \nu t$. Every orbit returns to its starting point at $t = 2\pi/\nu$.*

Counting

$N^2 - 1$ constraint equations, and a residual gauge group of dimension $1 + (N - 1)^2$ whose scalars act trivially, leave $2N^2 - (N^2 - 1) - (N - 1)^2 = 2N$, as the slice says (Figure 12.4).

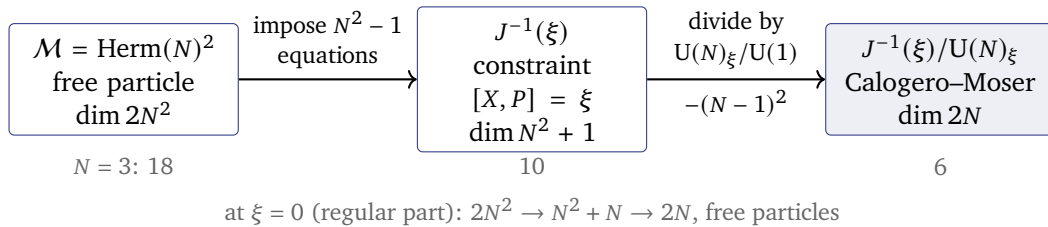


Figure 12.4: The reduction diagram: a free particle; the surface where the conserved $[X, P]$ takes one value; the quotient by what preserves that value. Only the non-zero level gives an interaction.

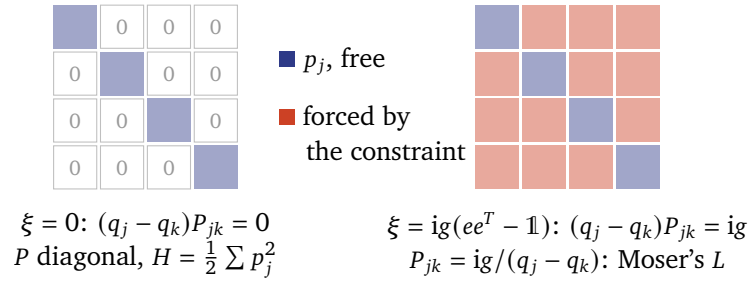


Figure 12.5: What survives of P once $X = Q$ is diagonal. The constraint fixes every off-diagonal entry: at level zero to 0 (free particles), at the level (12.4) to $ig/(q_j - q_k)$ (Calogero–Moser).

12.6 Why the Lax pair was inevitable

Read [Theorem 12.12](#) as a dictionary. Moser's L is the momentum matrix P in the gauge where X is diagonal; since P is conserved, its gauge-fixed image $L = U^\dagger P U$ changes only by conjugation, and *isospectrality* is the conservation of the free momentum. Moser's M is minus the velocity $U^\dagger \dot{U}$ of the gauge transformation needed to keep X diagonal. The integrals $\text{tr } L^k = \text{tr } P^k$ are functions of P alone, which Poisson-commute on $\mathcal{M}$ trivially; [Theorem 12.7](#) passes this to the reduced space, so the involution that [Theorem 11.6](#) needed a dynamical r -matrix for is, from here, automatic. The asymptotic momenta of § 11.5 are the eigenvalues of P , and § 11.5's perturbation argument for zero displacement now rests on the proved formula (12.6). And $\mathfrak{sl}_N$ enters because the constraint $(q_j - q_k)P_{jk} = ig$ is indexed by the differences $q_j - q_k$, the roots of $\mathfrak{sl}_N$ (reading, below).

Part I in one page. The classical structures of Part I, and what Part II makes of each:

Part I	built in	becomes, in Part II
$\{f, g\}, \dot{f} = \{f, H\}$, first integrals	Lecture 1	$[\cdot, \cdot]/i\hbar$; what counts as integrable, Lecture 13
Liouville–Arnold, with its hypotheses $\oint p \cdot dq = 2\pi I$	Lecture 4 Lecture 5	the question Lecture 13 asks quantisation conditions; Bethe equations, Lecture 15
Lax pair, $\text{tr } L^k$ conserved	Lecture 7	the Lax operator $L_n(\lambda)$, Lecture 17
Sklyanin bracket, $\{t(\lambda), t(\mu)\} = 0$, CYBE	Lectures 8 and 10	RTT , commuting transfer matrices, YBE, Lecture 16 , Lecture 17
momenta permuted, shifts order-independent	Lecture 9	factorised scattering, Lecture 16
Calogero–Moser by reduction	Lectures 11 and 12	quantum Calogero–Sutherland, Lecture 21

The first six rows are the export contract; the last is this lecture's, and [Lecture 21](#) quantises it. Reduction itself — free motion, a symmetry, a constraint, a gauge fixing — is one of the construction principles [Lecture 23](#)'s frontier returns to.

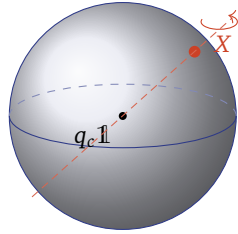
12.7 Two by two

Everything in this lecture can be seen for $N = 2$, in the coordinates $X = q_c \mathbb{1} + \mathbf{x} \cdot \boldsymbol{\sigma}$ of [Example 12.1](#).

Orbits. The orbit of a regular X is the sphere $|\mathbf{x}| = r$ at fixed q_c , of dimension $2 = 4 - 2$ ([Proposition 12.3](#)); conjugation rotates $\mathbf{x}$, and the stabiliser rotates about the axis through $\mathbf{x}$ ([Figure 12.6](#)). A degenerate $X = q_c \mathbb{1}$ is its own orbit, a single point, with all of $U(2)$ as stabiliser: the bigger the stabiliser, the smaller the orbit, $\dim(\text{orbit}) = \dim G - \dim(\text{stabiliser})$.

The momentum map. $[\mathbf{x} \cdot \boldsymbol{\sigma}, \mathbf{v} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{x} \times \mathbf{v}) \cdot \boldsymbol{\sigma}$, so $J = 2i\boldsymbol{\ell} \cdot \boldsymbol{\sigma}$ and $\xi = ig\sigma_1$ is $\boldsymbol{\ell} = (g/2, 0, 0)$. Its stabiliser is $\{U : Ue \propto e\}$, the unitaries diagonal in the eigenbasis of σ_1 : rotations about the ℓ -axis, $U(1) \times U(1)$, of dimension $2 = 1 + (N - 1)^2$. The orbit of ξ is the sphere $|\boldsymbol{\ell}| = g/2$, of dimension $2N - 2 = 2$ ([Proposition 12.8](#)).

The reduction. $8 = 2N^2$ dimensions; the constraint fixes the three components of $\boldsymbol{\ell}$, leaving 5; the residual rotations about $\boldsymbol{\ell}$ remove 1, leaving $4 = 2N$: centre of mass, gap, and their momenta. Gauge-fixing $\mathbf{x}$ onto the σ_3 axis makes X diagonal, and the constraint then forces $\mathbf{v}$'s σ_2 -component to carry the angular momentum: $P_{12} = ig/(q_1 - q_2)$.



orbit of X : the sphere $|\mathbf{x}| = r$,
same eigenvalues $q_c \pm r$

stabiliser of X : rotations about
the axis through X

Figure 12.6: An orbit of conjugation, $N = 2$, at fixed trace. Hermitian matrices with eigenvalues $q_c \pm r$ form a sphere of radius r ; the diagonal ones are its two poles on the σ_3 axis. The stabiliser of a point acts as the rotations about the axis through it (with the scalars, the unitaries diagonal in a basis where the point is diagonal). The centre, $r = 0$, is an orbit on its own.

12.8 Regular and singular levels

Why ξ is a regular value of J and 0 is not. For a tangent vector v at a point of $\mathcal{M}$, $\text{tr}(A dJ(v)) = dJ_A(v) = \omega(X_{J_A}, v)$, which vanishes for every v exactly when $X_{J_A} = 0$ there, i.e. when e^{sA} fixes the point. So the image of dJ is everything orthogonal to the Lie algebra of the point's stabiliser. On $J^{-1}(\xi)$, by [Theorem 12.11\(a\)](#), every point is conjugate to a slice point (Q, L) ; a unitary fixing Q is diagonal, and $DLD^\dagger = L$ with every $L_{jk} \neq 0$ forces $D_{jj} = D_{kk}$. Only the scalars fix the point, so dJ is onto the traceless anti-Hermitian matrices, of dimension $N^2 - 1$. At $(Q, \text{diag}(p)) \in J^{-1}(0)$ with Q regular the stabiliser is T^N , and the rank drops to $N^2 - N$. Near such a point $J^{-1}(0)$ consists of pairs diagonalised together, X still with distinct eigenvalues: N eigenvalues of X , N of P and an orbit $U(N)/T^N$ of dimension $N^2 - N$, in all $N^2 + N$, not $N^2 + 1$. (The same holds wherever the joint eigenspaces of X and P are lines, e.g. at $X = 0$ with P regular; X regular is merely the part [Proposition 12.10](#) uses.) At $(X, P) = (0, 0)$ the level set is not a manifold: it is a cone (invariant under $(X, P) \mapsto (sX, sP)$),

and a cone that is a manifold at its vertex is a linear subspace; this one is not, since $(\sigma_3, 0)$ and $(0, \sigma_1)$ lie on it and their sum does not.

Reading

12.9 Root systems and the other Calogero–Moser models

The differences $q_j - q_k$ are not an accident of notation. The diagonal matrices Q act on the matrix units by $[Q, E_{jk}] = (q_j - q_k)E_{jk}$: the numbers $q_j - q_k$ are the eigenvalues of ad_Q , which in Lie theory are the values at Q of the roots $e_j - e_k$ of $\mathfrak{sl}_N$ (the root system A_{N-1}). The constraint of [Theorem 12.11](#) is one equation per root space, and the Calogero–Moser potential is a sum over the positive roots α of $g^2/\alpha(q)^2$. For $N = 3$, in the plane $q_1 + q_2 + q_3 = 0$ where the centre of mass has been removed, the three positive roots are drawn in [Figure 12.7](#): each is a pair of particles, and its hyperplane $\alpha(q) = 0$ is where that pair would collide. The chamber $q_1 < q_2 < q_3$ of [Theorem 12.11](#) is a Weyl chamber, cut out by those walls.

Olshanetsky and Perelomov (1976) replaced A_{N-1} by other root systems R :

$$H = \frac{1}{2} |p|^2 + \sum_{\alpha \in R_+} g_\alpha^2 V(\alpha(q)),$$

with couplings g_α equal on roots related by the Weyl group. For BC_N (which contains B_N , C_N , D_N as special couplings) and V rational, hyperbolic or trigonometric, they constructed a Lax pair, and so proved Liouville integrability, *for couplings obeying one further relation*: with g on the roots $e_j \pm e_k$, g_1 on e_j and g_2 on $2e_j$, either $g_1 = 0$ or $g_1^2 + \sqrt{2} g_1 g_2 - 2g_2^2 = 0$ (Perelomov, §3.9, Proposition 2) · *cited*. The exceptional systems are integrable too, but Perelomov records no Lax pair for them. For B_N , C_N , D_N the positive roots include $e_j + e_k$ as well as $e_j - e_k$ (and e_j or $2e_j$), so particle j interacts also with the mirror image of particle k and, for B_N and C_N , with a wall at the origin. For those couplings they obtained the rational and hyperbolic models by projecting free motion on symmetric spaces of zero and negative curvature, and the trigonometric one from positive curvature — the construction of this lecture, with $\text{Herm}(N)$ one example (Perelomov, §3.3). In the same spirit, Kazhdan, Kostant and Sternberg obtained the trigonometric model $1/\sin^2$ by replacing the matrix X with a unitary matrix, so that its eigenvalues live on a circle · *cited*. Root systems return in [Lecture 22](#), as the Dynkin diagrams indexing nested Bethe equations, where the rep-engine package supplies the representation theory.

Reading

12.10 Ruijsenaars–Schneider

Ruijsenaars and Schneider (1986) found a one-parameter deformation of every Calogero–Moser model, still integrable, that behaves like a relativistic particle system: with c a “speed of light” and unit masses, the rational one is

$$H_{\text{RS}} = c^2 \sum_{j=1}^N \cosh \frac{p_j}{c} \prod_{k \neq j} \left(1 + \frac{g^2}{c^2 (q_j - q_k)^2} \right)^{1/2} = Nc^2 + H_{\text{CM}} + O(c^{-2})$$

(Arutyunov, (1.67); the expansion is checked in `scripts/check-112-algebra.py`). It too comes from reduction: the rational model from the cotangent bundle of the group $\text{GL}(N)$ in

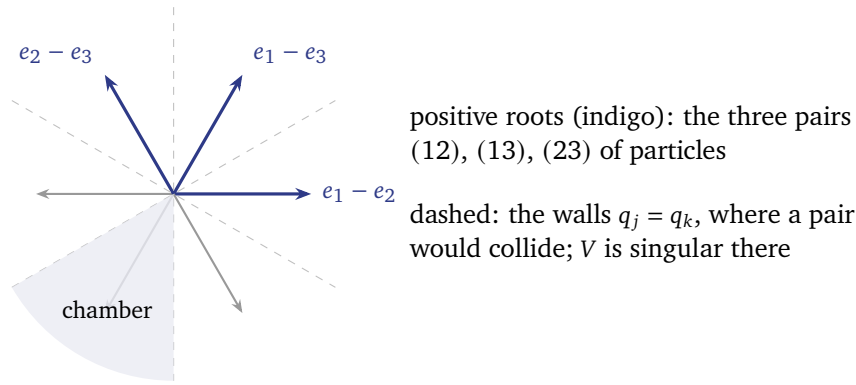


Figure 12.7: The root system A_2 of $\mathfrak{sl}_3$, drawn in the plane $q_1 + q_2 + q_3 = 0$. Each positive root is one interacting pair; the six walls bound the Weyl chambers, and the particles stay in one of them. (The chamber is shaded only schematically; its sides are two of the walls.)

place of the matrix space (Arutyunov, §2.2.1), and the hyperbolic one from a Poisson–Lie deformation of that, the “Heisenberg double” (§2.2.2) · *cited*. This course does not develop either. The elliptic case is Ruijsenaars (1987). Its quantum version is beyond this course.

Summary.

- *Free motion* on $\mathcal{M} = \text{Herm}(N)^2$, $\omega = \text{tr}(dX \wedge dP)$, $H = \frac{1}{2} \text{tr} P^2$: $X(t) = X_0 + tP_0$ ((12.3)).
- *Conjugation* by $U(N)$: orbits are isospectral sets; a regular X has stabiliser the diagonal torus (Proposition 12.3).
- *The momentum map is a commutator*, $J = [X, P]$, conserved and equivariant (Proposition 12.5 and Corollary 12.6); for $N = 2$ it is the angular momentum.
- *Only the stabiliser acts on a level set*: for $\xi = \text{ig}(ee^T - \mathbf{1})$ it is $\{U : Ue \propto e\}$ (Proposition 12.8), and on $J^{-1}(\xi)$ the eigenvalues of X cannot collide (Lemma 12.9).
- *Reduction twice*: at 0, free particles (Proposition 12.10); at ξ , Calogero–Moser, with P gauge-fixed to Moser’s L and canonical (q, p) (Theorem 12.11). Dimensions $2N^2 \rightarrow N^2 + 1 \rightarrow 2N$.
- *The positions are the eigenvalues of $Q(0) + tL(0)$* , and Moser’s M is minus the gauge velocity (Theorem 12.12); in the trap, of $Q(0) \cos \nu t + \nu^{-1}L(0) \sin \nu t$ (Corollary 12.13).

Tutorial. Exercise Sheet 12.

References

D. Kazhdan, B. Kostant and S. Sternberg, *Comm. Pure Appl. Math.* 31 (1978) 481 (the reduction); Perelomov, *Integrable Systems of Classical Mechanics and Lie Algebras*, §1.9 (the projection method) and §3.3 (free motion on Hermitian matrices, eqs. 3.3.1–3.3.11); P. Etingof, *Lectures on Calogero–Moser systems*, §§1.4, 2.5–2.7 (reduction along an orbit, the complex version); J. Marsden and A. Weinstein, *Rep. Math. Phys.* 5 (1974) 121; A. Cannas da Silva, *Lectures on Symplectic Geometry*, Thm 23.1 and §24.4; G. Arutyunov, *Elements of Classical and Quantum Integrable Systems*, §2.1.2 (constraints and gauge), §2.2 (Calogero–Moser and Ruijsenaars–Schneider from reduction); M. A. Olshanetsky and A. M. Perelomov,

Invent. Math. 37 (1976) 93; S. N. M. Ruijsenaars and H. Schneider, Ann. Phys. 170 (1986) 370; S. N. M. Ruijsenaars, Commun. Math. Phys. 110 (1987) 191.